

Group-Theoretic Constructions

There are two major ways we may construct new groups out of old: extensions/semidirect products and pushouts/amalgamated free products. We will provide a brief overview of both of these.

Definition: Let Q and N be groups. We say a group G is an extension of Q and N if there is an exact sequence

$$1 \rightarrow N \rightarrow G \rightarrow Q \rightarrow 1.$$

If the extension splits — i.e., there is a section $Q \rightarrow G$ — then $G \cong Q \rtimes_{\varphi} N$ is a semidirect product for some $\varphi: Q \rightarrow \text{Aut}(N)$.

Example:

- If the homomorphism $\varphi: Q \rightarrow \text{Aut}(N)$ is trivial, then $N \rtimes_{\varphi} Q \cong N \times Q$ is a direct product.
- The map $\mathbb{Z}/2\mathbb{Z} \rightarrow \text{Aut}(\mathbb{Z}/n\mathbb{Z})$ given by inversion yields the dihedral group

$$D_n \cong \mathbb{Z}/2\mathbb{Z} \rtimes \mathbb{Z}/n\mathbb{Z}.$$

- If G is any group, the lamplighter group of G is the semidirect product $\mathbb{Z} \rtimes G^{\mathbb{Z}}$ with homomorphism $\varphi: \mathbb{Z} \rightarrow \text{Aut}(G^{\mathbb{Z}})$ given by

$$\varphi(z) = ((g_n)_{n \in \mathbb{Z}} \mapsto (g_{n+z})_{n \in \mathbb{Z}}).$$

- the wreath product of G and H , denoted $G \wr H$, is the semidirect product $H \rtimes G^H$ with $\varphi: H \rightarrow \text{Aut}(G^H)$ given by the shift action

$$\varphi(h) = ((g_k)_{k \in H} \mapsto (g_{kh})_{k \in H}).$$

Groups acting on Graphs

Quasi-Isometry of Metric Spaces and Groups

Growth types of Groups

Group Homology/Cohomology

Amenability of Groups

References

- [Löh17] Clara Löh. *Geometric Group Theory*. Universitext. An Introduction. Springer, Cham, 2017, pp. xi+389. ISBN: 978-3-319-72253-5; 978-3-319-72254-2. DOI: [10.1007/978-3-319-72254-2](https://doi.org/10.1007/978-3-319-72254-2). URL: <https://doi.org/10.1007/978-3-319-72254-2>.
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